

# LECTURE – CONSTRUCTOR OOP (JAVA)



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# AGENDA



- Constructor
- Types of constructor
- Constructor overloading
- Examples

# CONSTRUCTOR



- In **Java**, a constructor is a block of codes similar to the method.
- It is called when an instance of the **class** is created.
- At the time of calling constructor, memory for the object is allocated in the memory.
- It is a special type of method which is used to initialize the object.
- Every time an object is created using the `new()` keyword, at least one constructor is called.
- It calls a default constructor if there is no constructor available in the class.
- In such case, Java compiler provides a default constructor by default.

# CONSTRUCTOR



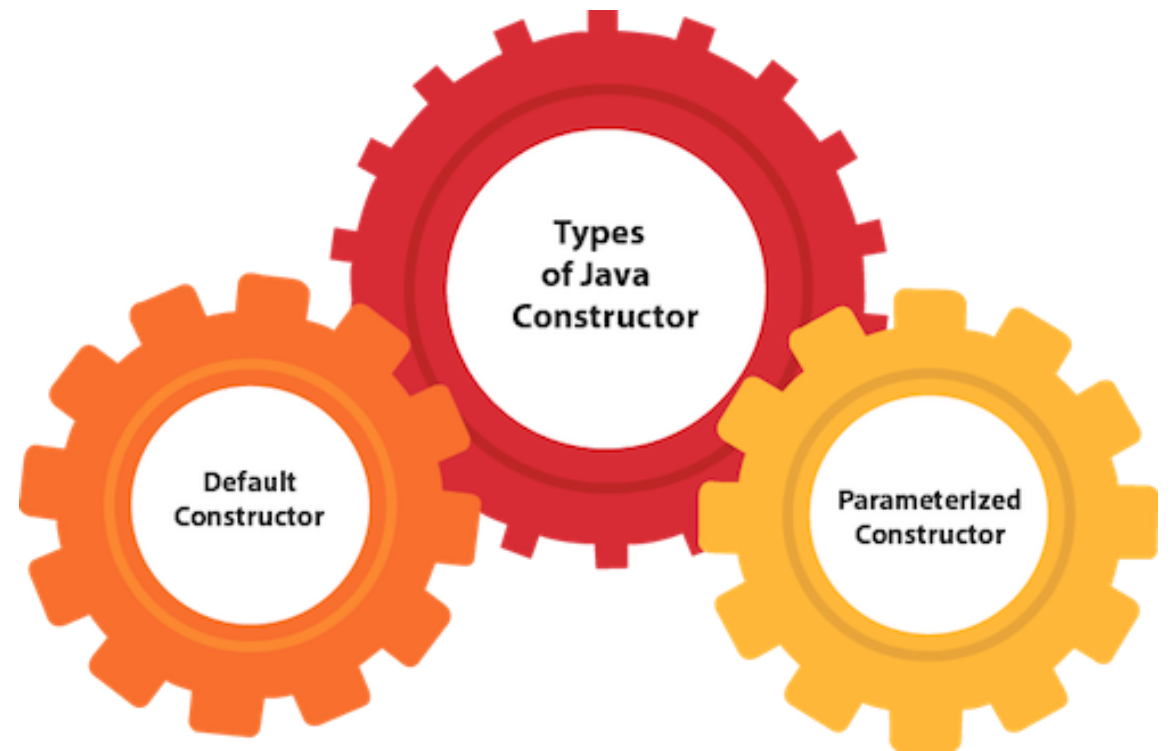
- There are two types of constructors in Java: no-arg constructor, and parameterized constructor.
- **Note:** It is called constructor because it constructs the values at the time of object creation. It is not necessary to write a constructor for a class. It is because java compiler creates a default constructor if your class doesn't have any.

# RULES FOR CREATING JAVA CONSTRUCTOR

- There are two rules defined for the constructor.
  1. Constructor name must be the same as its class name
  2. A Constructor must have no explicit return type
  3. A Java constructor cannot be abstract, static, final, and synchronized

# TYPES OF JAVA CONSTRUCTORS

- There are two types of constructors in Java:
  1. Default constructor (no-arg constructor)
  2. Parameterized constructor



# DEFAULT CONSTRUCTOR

- Constructor is called default constructor when it does not have any value.

```
class Bike1{  
    Bike1(){  
        System.out.println("Bike is new");  
    }  
    public static void main(String args[]){  
        Bike1 b=new Bike1();  
    }  
}
```

- In this example, we are creating the no-arg constructor in the Bike class. It will be invoked at the time of object creation

# JAVA PARAMETERIZED CONSTRUCTOR

- A constructor which has a specific number of parameters is called a parameterized constructor.
- **Why use the parameterized constructor?**
- The parameterized constructor is used to provide different values to distinct objects. However, you can provide the same values also.
- In this example, we have created the constructor of Student class that have two parameters. We can have any number of parameters in the constructor.

```
class Student4{  
    int id;  
    String name;  
    Student4(int i,String n){  
        id = i;  
        name = n;  
    }  
    void display()  
{System.out.println(id+" "+name);}  
    public static void main(String args[]){  
        Student4 s1 = new Student4(111,"Karan");  
        Student4 s2 = new Student4(222,"Aryan");  
        s1.display();  
        s2.display();  
    }  
}
```



# CONSTRUCTOR OVERLOADING IN JAVA

- In Java, a constructor is just like a method but without return type. It can also be overloaded like Java methods.
- Constructor overloading in Java is a technique of having more than one constructor with different parameter lists. They are arranged in a way that each constructor performs a different task. They are differentiated by the compiler by the number of parameters in the list and their types.

Thanks 😊